Traffic Speed Forecasting with Spatio-Temporal Graph Neural Networks Incorporating Public Transport Demand

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Abstract

Accurate traffic speed estimation is central to urban navigation systems. However, most existing prediction models rely solely on road traffic data and neglect interactions between road traffic and public transport (PT) demand, which can reflect demand fluctuations and congestion effects.

In this work, we model the urban street network as a graph and investigate the benefits of incorporating PT demand into traffic speed forecasting using four established spatio-temporal graph neural network frameworks. Road traffic speed and PT usage are combined as dynamic node features within these GNN architectures, while road connectivity is represented by a fixed adjacency matrix. Beyond standard train–test evaluation, we analyze whether the inclusion of PT demand improves model robustness and reduces the need for frequent retraining under evolving traffic conditions. The proposed approach is evaluated on real-world data from Bogotá, Colombia, a highly congested metropolitan area.

1 Introduction

Urban traffic can be modeled as a graph, where nodes and edges represent the city's layout and time-varying node features capture evolving traffic conditions. Traffic dynamics are strongly influenced by disruptions in other transportation modes. Events such as strikes, delays, or capacity reductions in public transport (PT) can induce modal shifts, leading commuters to switch to private vehicles [13]. This often exacerbates road congestion and results in decreased traffic speeds [18]. While explicit disruption indicators are typically unavailable in real-world datasets, their effects are reflected in observable variations in PT usage, such as passenger counts.

Many navigation systems rely on shortest path prediction [2], which requires accurate estimates of traffic speed or travel time. As such estimations usually ignore dynamic PT usage patterns and their direct impact on road traffic, the core challenge addressed in this work is the accurate estimation of traffic speeds for private vehicles by adding contextual data. To this end, we employ four established Spatio-Temporal Graph Neural Network (ST-GNN) architectures: STGCN, DCRNN, MTGNN, and Graph WaveNet [10, 23–25]. The selected models represent architecturally diverse spatio-temporal modeling paradigms, allowing us to assess whether the integration of PT demand generalizes across different ST-GNN designs. In addition to predictive accuracy, maintaining model performance over time is a key challenge, as evolving traffic patterns can lead to performance degradation and necessitate periodic retraining [8].

Effectively combining heterogeneous data sources remains challenging. Most existing approaches rely primarily on road traffic data, sometimes augmented with weather or other contextual factors [6, 14, 16, 26]. To the best of our knowledge, PT load information has not been systematically integrated into graph-based traffic speed prediction models, leaving the joint modeling of road traffic and PT demand largely unexplored, partly due to the scarcity of synchronized datasets for the same city and time period.

We conduct our analysis using data from Bogotá, Colombia, a megacity consistently ranked among the most congested cities worldwide [15]. Our analysis builds on synchronized real-world datasets, containing average road segment speeds and PT boarding and alighting counts, both available at a 15-minute resolution. We hypothesize that PT demand serves as an early signal of modal shifts, improving predictive accuracy and robustness under evolving traffic conditions. This work makes the following contributions:

- We evaluate four architecturally diverse ST-GNN architectures (STGCN, DCRNN, MTGNN, and Graph WaveNet) on a synchronized 15-minute resolution dataset from Bogotá that combines road traffic speeds with interpolated public transport counts in a unified graph setting, and provide a quantitative comparison against a traffic-only model.
- We study how the inclusion of public transport demand affects the need for periodic retraining of these ST-GNN models under evolving traffic conditions.

2 Related Work

GNN Models for Traffic Prediction. ST-GNNs are widely used for traffic forecasting due to their ability to jointly model spatial dependencies on road networks and temporal traffic dynamics [11]. In this work, we focus on established ST-GNN architectures. Early convolutional approaches such as STGCN combine graph convolutions with standard temporal convolutions to capture local spatial and temporal patterns on fixed graph structures [25]. Building on this paradigm, Graph WaveNet employs dilated temporal convolutions to efficiently expand the temporal receptive field and model long-range temporal dependencies, while primarily operating on a predefined and optionally adaptive graph [17, 24]. In contrast, MTGNN does not require predefined network connectivity and instead learns latent graph structures directly from multivariate traffic time series, enabling the capture of spatial correlations beyond physical road topology [23]. DCRNN represents a recurrent modeling approach that combines diffusion-based graph convolutions on directed graphs with sequence-to-sequence recurrent neural networks to model traffic propagation along road networks [10]. These models represent key architectural directions in spatio-temporal traffic prediction.

Using Additional Data for Traffic Prediction. Incorporating additional data sources, such as social events, weather conditions, real-time traffic sensor data, and historical traffic patterns, improves the performance of traffic prediction models [1, 27]. In particular, weather data has been widely incorporated into traffic prediction models and has been shown to improve prediction performance [14, 16, 26]. Beyond weather data, other contextual

factors such as points of interest (POIs) and social influences (e.g., public holidays or major events) have also been shown to affect traffic conditions and prediction accuracy [6, 28]. However, PT demand is typically not considered as an explicit input in these models, despite its strong interaction with road traffic conditions.

Influence of Model Retraining on Traffic Prediction. The assumption that the training data adequately represents all future test data may not hold in practice (e.g., due to concept drift), which can result in model performance degradation [3]. Studies have investigated model retraining as a mechanism to address non-stationarity in traffic data and to sustain forecasting performance over time [8]. However, these analyses focus primarily on traffic-only models and do not consider how additional contextual information, such as public transport demand, may influence retraining requirements.

3 Experimental Apparatus

3.1 Datasets

We integrate multiple data sources capturing road traffic conditions, PT usage, and urban infrastructure in Bogotá, Colombia.

Road traffic speed data. Traffic speed data were obtained from the open data portal of the Bogotá District Secretariat of Mobility and are collected via the Bitcarrier system, which estimates vehicle speeds from anonymized Wi-Fi and Bluetooth signals. The dataset provides average vehicle speeds per road segment at 15-minute intervals and covers the period from February 2019 to November 2022 [4, 12].

PT load data and stop locations. PT data were obtained from Bogotá's open data portal and consist of aggregated passenger boarding and alighting counts per stop and time interval, derived from automated fare validation within the TransMilenio system [20, 21]. The data is available from August 2012 to November 2025. A separate dataset maps PT stop identifiers to their geographic coordinates [19].

Summary. Together, these datasets enable a joint spatio-temporal representation of road traffic speeds and PT demand. Although the datasets overlap from February 2019 to November 2022, we restrict our analysis to the period from February 2019 to March 2020 to exclude the structural disruptions caused by COVID-19 lockdown measures starting end of March 2020 in Bogotá [7, 22]. Both datasets are reported at 15-minute intervals, enabling direct comparability.

3.2 Preprocessing

To integrate traffic speed and PT load data into a unified representation, we model the street network of Bogotá as a graph $G = (V, E)$. Each node $v \in V$ represents a road segment as defined in [12]. An edge $e \in E$ connects two nodes if the corresponding segments share a common start or end point. PT counts are mapped onto the road network by computing the midpoint of each road segment and spatially interpolating both passenger alighting and boarding counts from nearby bus stops. For each node v and time t, boarding and alighting loads are interpolated separately using inverse distance weighting (IDW), as in [14]:

$$\ell_v^t = \frac{\sum_{s \in \mathcal{S}_v^k} w_{vs}\, \ell_s^t}{\sum_{s \in \mathcal{S}_v^k} w_{vs}}, \qquad w_{vs} = \frac{1}{d(m_v, s)^p}, \tag{1}$$

where $\mathcal{S}_v^k$ denotes the set of the k nearest bus stops to the segment midpoint m_v, ℓ_s^t is the number of people alighting/boarding at stop s, $d(m_v, s)$ is the Euclidean distance, and p controls the distance decay. This yields a time series of node-level feature vectors x_v^t for each node, consisting of traffic speed measurements for both travel directions and interpolated PT boarding and alighting counts. Nodes with excessive missing data are removed during preprocessing, yielding a fixed road network graph that is used for all subsequent time steps. Extreme outliers are handled using percentile-based clipping. Following [11], node features are standardized using z-score normalization to account for differing scales between traffic speed and passenger counts [5]. Table 1 shows the descriptive statistics of the dataset after preprocessing. Calendar effects are deliberately excluded to isolate the contribution of PT demand, as they capture recurring demand patterns that could otherwise confound this assessment.

Table 1: Descriptive statistics of the Bogotá road speed and PT dataset after preprocessing (February 2019 to March 2020).

Property	Value
Time resolution	15 minutes
Number of timestamps	≈ 38,000
Number of nodes	≈ 700
Number of edges	≈ 1,200
Average nearby PT stops per node	k

3.3 Procedure

Model setup and comparison. We employ four established ST-GNN architectures (namely STGCN, DCRNN, MTGNN and Graph WaveNet) to forecast traffic speeds on road segments. The chosen models represent different spatio-temporal modeling philosophies (e.g., diffusion-based, convolutional, adaptive graph learning), allowing us to assess whether the observed effects consistently hold across fundamentally different ST-GNN designs. For each architecture, a traffic-only baseline model and a second variant incorporating interpolated PT counts are trained and evaluated using an identical procedure to enable direct comparison. Based on [14], the models are trained for short-term traffic speed forecasting using a window of the past 24 hours as input. Traffic speeds are predicted for forecasting horizons of 15, 30, and 60 minutes ahead, consistent with the evaluation protocol described in [11]. The dataset is chronologically split into training, validation, and test sets in a 7:1:2 ratio, following common practice summarized in [11].

Retraining analysis. To assess the impact of PT demand on retraining requirements, models are evaluated with and without periodic retraining after an initial training phase. Fixed retraining intervals of one, two, and three months are considered by shifting a fixed-length training window forward in time and retraining the models from scratch on chronologically subsequent observations, alongside a no-retraining baseline.

3.4 Hyperparameter Optimization

Hyperparameters are selected via grid search on the validation set for each ST-GNN architecture. Tuned model parameters include the learning rate, number of hidden units, batch size, and number of training epochs. For PT load interpolation, the number of nearest stops k and the distance decay parameter p (commonly set to 1 or 2 [14]) of the inverse distance weighting scheme are also optimized using grid search over predefined ranges.

3.5 Metrics

Model performance is evaluated using the Mean Absolute Error (MAE), the Root Mean Squared Error (RMSE), and the Mean Absolute Percentage Error (MAPE), three standard regression metrics for traffic speed forecasting [9, 11].

Use of Generative AI Tools

Portions of this proposal were refined using generative AI tools for language editing and paraphrasing. All research questions, methodological framework, proposed evaluation strategy, and scientific reasoning were developed independently by the author.

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